Hydrodynamic Phase-Crossover and the Origin of the Pseudogap in Constrained Many-Body Fluids

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Abstract

We present a formal derivation of the transition from microscopic kinematic constraints to collective hydrodynamic stiffness in two-dimensional correlated electron systems. We demonstrate that the pseudogap regime in the Fermi-Hubbard model represents a phase-crossover into a constrained quantum fluid where single-particle excitations are suppressed by a hydrodynamic backflow mechanism. Utilizing the principle of delayed synchronization in closed circulation loops, we show that discrete energy levels emerge as stable phase-locked states of the collective field. We calibrate the crossover using a phenomenological transport velocity $v \sim 10^6$ m s^{-1}, consistent with experimentally observed Dirac and spinon velocities in correlated two-dimensional materials, and evaluate the resulting spectral weight suppression.

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1 Effective Action for a Constrained Medium

In the strong-coupling limit of the Hubbard model ($U \gg t$), the suppression of double occupancy restricts the dynamics to a manifold of constrained many-body states [2]. In the continuum limit, the low-energy physics is governed by collective fields representing charge density fluctuations $\rho(\mathbf{x}, t)$ and a local orientation field $\mathbf{S}(\mathbf{x}, t)$ associated with the spin-density background.

The effective Euclidean action for this constrained fluid is given by:

$$S_{\text{eff}} = \int d^2x\, dt \left[\frac{\chi_\rho}{2}(\partial_t \rho)^2 + \frac{\chi_S}{2}(\partial_t \mathbf{S})^2 + \mathcal{F}_{\text{hydro}}[\rho, \mathbf{S}]\right] \tag{1}$$

where the hydrodynamic free energy functional $\mathcal{F}_{\text{hydro}}$ accounts for the stiffness of the collective modes:

$$\mathcal{F}_{\text{hydro}} = \int d^2x \left[\frac{K_\rho}{2}(\nabla \rho)^2 + \frac{K_S}{2}(\nabla \mathbf{S})^2 + \lambda\, \rho\, \nabla \cdot \mathbf{S}\right] \tag{2}$$

Physically, the functional $\mathcal{F}_{\text{hydro}}$ penalizes spatial inhomogeneities in the constrained medium, so that at low temperatures the dynamics is dominated by collective, stiffness-controlled modes rather than independent quasiparticle motion. The term λ represents the spin–charge coupling induced by the kinematic constraints of the lattice [3]. Here, K_S denotes the collective stiffness that emerges as T drops below a characteristic crossover temperature T^*.

We emphasize that the present framework is intended as an effective description of strongly constrained two-dimensional electronic systems in the pseudogap regime, where single-particle quasiparticles are strongly suppressed and transport is dominated by collective response. While the Hubbard model provides the motivating example, no claim of universality across all correlated materials is made. Rather, the approach applies within parameter regimes where kinematic constraints generate an emergent collective stiffness.

2 Phase Locking and Quantization in Delayed Loops

Within this effective description, the stiffness parameters K_S, K_ρ and the susceptibilities χ_S, χ_ρ encode collective response rather than microscopic details. In principle, these quantities could be extracted from numerical approaches such as cluster dynamical mean-field theory, slave particle formulations, or density-matrix renormalization group calculations, where suppression of quasiparticle coherence and enhanced collective stiffness are well documented. The present work does not attempt such a derivation, but instead focuses on the generic dynamical consequences once collective stiffness is established.

Once collective stiffness has emerged, transport in the constrained fluid necessarily occurs at a finite velocity, implying that closed circulation paths experience delayed self-interaction rather than instantaneous response. The emergence of stiffness K_S facilitates a characteristic transport velocity $v = \sqrt{K_S/\chi_S}$. For a closed circulation loop of length L, the finite propagation time $\tau = L/v$ introduces a delayed self-interaction in the field dynamics. Such systems are known to exhibit discrete mode selection via phase-locking [4].

2.1 Geometric Estimate and Mode Spacing

To assess whether the delay-induced quantization occurs at physically relevant energy scales, it is useful to estimate the characteristic loop length and associated propagation time. Consider a representative circulation loop of length $L \sim \mathcal{O}(a)$, spanning a few lattice constants. For $L \approx 4 \times 10^{-10}$ m and a phenomenological transport velocity $v \approx 10^6$ m s^{-1}, the round-trip time is:

$$\tau = \frac{L}{v} \approx 4 \times 10^{-16} \text{ s}. \tag{3}$$

The phase θ of the collective excitation obeys a delayed synchronization law $\dot{\theta}(t) = \Omega - \kappa \sin(\theta(t) - \theta(t-\tau))$. Stationary, phase-locked solutions require the phase closure condition $\oint_{\mathcal{C}} \nabla\theta \cdot d\mathbf{l} = 2\pi n$. This

implies a discrete energy spacing $\Delta E \approx h/\tau \sim 10$ eV, which sets an upper energy scale for coherent collective excitations in the stiffness regime.

In realistic systems, the circulation paths underlying the delayed feedback mechanism are expected to exhibit a distribution of lengths and to reflect lattice anisotropies or disorder. Such effects would broaden the collective mode spectrum and soften sharp quantization, but do not remove the characteristic energy scale set by the propagation delay. The phase-locking mechanism therefore survives statistically even in non-ideal geometries.

3 Relation to Competing Pseudogap Scenarios

It is instructive to contrast this hydrodynamic crossover mechanism with more conventional explanations of the pseudogap, which typically retain the quasiparticle as the primary degree of freedom. Standard scenarios generally fall into two categories:

1. **Preformed Pairs:** Postulates the formation of incoherent Cooper pairs above T_c.
2. **Competing Order:** Attributes the gap to a hidden symmetry-breaking phase (e.g., charge density waves).

In contrast, the constrained-fluid description requires neither preformed pairs nor symmetry breaking. While Cluster Dynamical Mean-Field Theory (CDMFT) and spin-fermion models [6] rely on the scattering of single-particle quasiparticles off fluctuations, the present framework treats the quasiparticle as a suppressed degree of freedom, with transport governed entirely by collective hydrodynamic response functions.

4 Numerical Results: Spectral Weight Suppression

The transition into the pseudogap is characterized by the transfer of spectral weight from single-particle Green's functions to incoherent collective modes. We define the renormalization factor Z through the hydrodynamic self-energy Σ_{hydro} arising from the backflow coupling λ [5]:

$$Z^{-1} = 1 - \left.\frac{\partial \text{Re}\Sigma_{\text{hydro}}}{\partial \omega}\right|_{\omega=0} \tag{4}$$

Assuming a characteristic velocity $v \approx 1.09 \times 10^6$ m s^{-1}, we evaluate the suppression as a function of $K_S(T)$. For $T < T^*$, we find $Z \approx 0.08$, indicating that 92% of the spectral weight is incoherent, matching the loss of coherence observed in the pseudogap phase.

Experimentally, the hydrodynamic crossover described here would manifest as a strong suppression of quasiparticle residue in angle-resolved photoemission spectroscopy, accompanied by a transfer of spectral weight into incoherent background features, consistent with the emergence of Fermi arcs in the pseudogap regime. Optical conductivity measurements would similarly reflect a redistribution of low-frequency spectral weight, while probes sensitive to collective excitations, such as Raman or neutron scattering, may detect broad, stiffness-controlled modes rather than sharp quasiparticle peaks.

5 Variational Scaling and Summary

Although microscopic details vary across materials, the variational structure of the hydrodynamic action constrains the scaling form of the low-lying excitations. For confined geometries in 2D, the variational principle applied to the hydrodynamic action yields a scaling of excitation energies:

$$\Delta E_n \propto \sqrt{K_S(T)\, n/\chi_S} \tag{5}$$

This $\sqrt{n}$ scaling provides a testable signature of the collective hydrodynamic nature of the state, distinct from the $1/n^2$ scaling of unconstrained systems.

Taken together, these results support the interpretation of the pseudogap as a hydrodynamic phase-crossover driven by kinematic constraints and collective stiffness, rather than as a precursor to symmetry breaking or pairing.

A Speculative Bounds and Temporal Jitter

We stress that the temporal jitter scale σ_τ introduced in this appendix is not required for the hydrodynamic crossover mechanism developed in the main text, nor does it enter any of the scaling results or physical conclusions. It is presented solely as an optional conjectural bound that may provide a possible falsifiability criterion, and its absence would not affect the validity of the collective phase-crossover picture.

References

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A Topology-modulated inertial mass functional

We use a factorized effective rest-mass functional,

$$M(T) = \mathcal{K}(T)\,\frac{u\,V(T)}{c^2}, \qquad u = \frac{1}{2}\rho_{\text{core}}\|\mathbf{v}_{\circlearrowleft}\|^2, \qquad V(T) \simeq \pi r_c^3 L_{\text{tot}}(T). \tag{6}$$

This yields the compact evaluation form

$$M(T) = M_0\, L_{\text{tot}}(T)\, k(T)^{-3/2}\, \varphi^{-g(T)}\, n(T)^{-1/\varphi} \left(\frac{4}{\alpha}\right)^{\sigma(T)}, \qquad M_0 = \frac{\pi r_c^3 \left(\frac{1}{2}\rho_{\text{core}}\|\mathbf{v}_{\circlearrowleft}\|^2\right)}{c^2}. \tag{7}$$

Here k is a spectral/complexity index, g a genus proxy, and n the number of link components. The exponent $\sigma(T) \in \{0, 1\}$ encodes whether "vacuum coupling" is effectively shielded ($\sigma = 0$) or exposed ($\sigma = 1$).

A.1 A rigorous shielding classifier for double twist knots

For the double twist family $K_{m,n}$, a sharp rational-sliceness classification is available [1]. We define the shielding exponent

$$\sigma(K_{m,n}) = \begin{cases} 0, & \text{if } K_{m,n} \text{ is rationally slice,} \\ 1, & \text{otherwise,} \end{cases} \tag{8}$$

and use the theorem [1]:

$$K_{m,n} \text{ rationally slice} \iff \big(mn = 0\big) \text{ or } \big(n = -m\big) \text{ or } \big(n = -m \pm 1\big). \tag{9}$$

Outside the $K_{m,n}$ family, $\sigma(T)$ remains a discrete model label to be fixed by the chosen topology-to-particle assignment.

B Atom-mass bridge via SEMF with an SST modulation

To connect constituent masses to atomic masses, we use a standard bookkeeping identity

$$M_{\text{atom}}(Z,N) \approx ZM_p + NM_n + ZM_e - \frac{E_{\text{bind}}(Z,N)}{c^2}, \qquad A = Z + N. \tag{10}$$

As a minimal phenomenological bridge, take the Bethe–Weizsäcker (SEMF) structure with an optional SST modulation factor $\Xi(A)$:

$$E_{\text{bind}}^{\text{SST}}(Z,N) = a_V^{(0)}\,\Xi(A)\,A - a_S^{(0)}\,\Xi(A)^{-1/2}\,A^{2/3} - a_C\frac{Z(Z-1)}{A^{1/3}} - a_A\frac{(A-2Z)^2}{A} + \delta(A,Z). \tag{11}$$

Setting $\Xi(A) = 1$ reduces to standard SEMF; deviations can be tested via out-of-sample mass residuals.

C Bridge: delayed loop-time selection and the $k(T)^{-3/2}$ measure

In closed-loop delayed synchronization models, a configuration selects (or strongly weights) a discrete set of self-consistent phase-locked modes characterized by a loop time τ (or an equivalent spectral gap scale). A minimal field-theoretic interpretation of the knot-index $k(T) \in \mathbb{R}^+$ used in the SST-59 mass kernel is:

$$T_{\text{prop}}(T) \;\propto\; \tau(T), \qquad k(T) \;=\; \frac{T_{\text{prop}}(T)}{T_{\text{prop}}(T_{\text{ref}})} \;=\; \frac{\tau(T)}{\tau(T_{\text{ref}})}.$$

For a Gaussian sector with quadratic operator $H = -\nabla^2 + m^2$ in three spatial dimensions, the proper-time / heat-kernel representation yields a universal short-time weight $T_{\text{prop}}^{-3/2}$ (the $d/2$ exponent with $d = 3$). Hence the measure contributes

$$T_{\text{prop}}(T)^{-3/2} \;\propto\; k(T)^{-3/2},$$

which matches the $k(T)^{-3/2}$ prefactor used in SST-59 without requiring additional model-specific assumptions beyond (i) loop-time selection and (ii) a quadratic fluctuation sector.